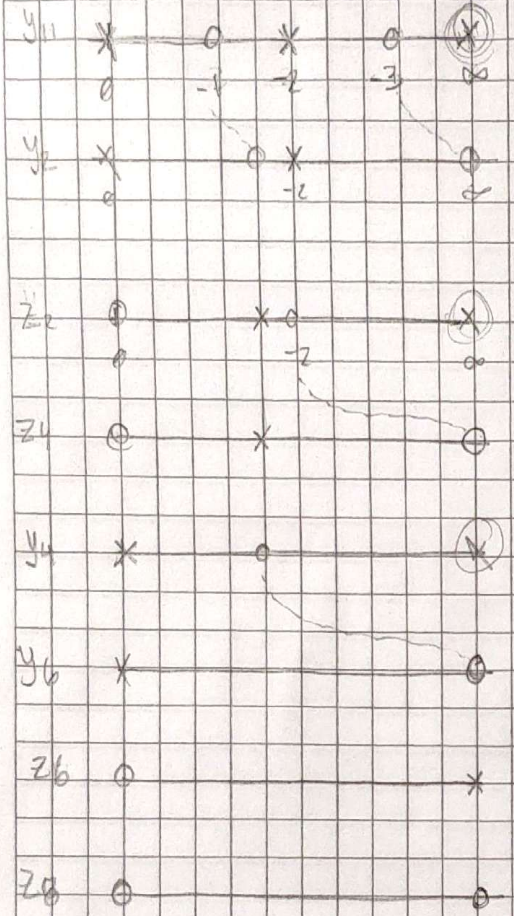
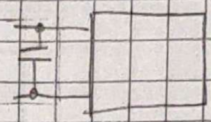
3) Sintetizar cuadripolo con $y_{21} = \frac{1}{s(s^2+4)}$ ∧ $y_{11} = \frac{(s^2+1)(s^2+9)}{s(s^2+4)}$

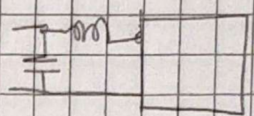
Sabiendo que $\frac{y_{21}}{y_{11}} = \frac{-I_2}{I}\Big|_{V_2=0} = \frac{\frac{1}{s(s^2+4)}}{\frac{(s^2+1)(s^2+9)}{s(s^2+4)}}$ con el aux $= s(s^2+4)$

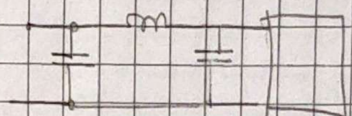


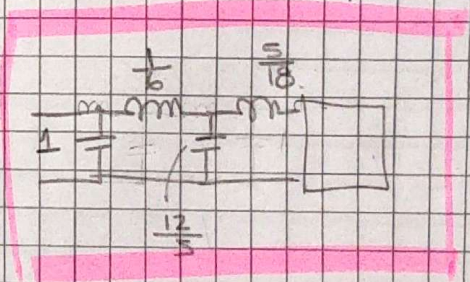
estamos sacando un polo en ∞



nuevamente polo en infinito pero en imp.







$\to y_2 = y_{11} - K_\infty \cdot s \Rightarrow \lim_{s\to\infty} \frac{y_{11}}{s} = \lim_{s\to\infty} \frac{(s^2+1)(s^2+9)}{s(s^2+4)s} = \lim_{s\to\infty} \frac{s^4+10s^2+9}{s^4+4s^2} = 1$

$y_2 = \frac{(s^2+1)(s^2+9)}{s(s^2+4)} - s = \frac{s^4+10s^2+9-(s^4+4s^2)}{s(s^2+4)} = \frac{6s^2+9}{s(s^2+4)}$

$\to z_2 = \frac{s(s^2+4)}{6s^2+9} \Rightarrow z_4 = z_2 - K_\infty \cdot s$ con $K_\infty = \lim_{s\to\infty} \frac{z_2}{s} = \lim_{s\to\infty} \frac{s(s^2+4)}{6(s^2+9)s} = \frac{1}{6}$

$z_4 = \frac{s(s^2+4)}{6s^2+9} - \frac{1}{6}s = \frac{\cancel{6s^3}+24s-\cancel{6s^3}-9s}{6(6s^2+9)} = \frac{5s}{12s^2+18}$

$\to y_4 = \frac{12s^2+18}{5s} \Rightarrow y_6 = y_4 - K_\infty \cdot s$ con $K_\infty = \lim_{s\to\infty} \frac{y_4}{s} = \lim_{s\to\infty} \frac{12s^2+18}{5s}\cdot\frac{1}{s} = \frac{12}{5}$

$y_6 = \frac{12s^2+18}{5s} - \frac{12}{5}s = \frac{\cancel{12s^2}+18-\cancel{12s^2}}{5s} = \frac{18}{5}s$

$\rightarrow Z6 = \frac{5 \cdot s}{10} \Rightarrow Z0 = Z6 - K_\infty s$ con $K_\infty = \lim_{s \to \infty} \frac{Z6}{s} = \lim_{s \to \infty} \frac{5 \cancel{s}}{10 \cdot \cancel{s}} = \frac{5}{10}$

$Z0 = \frac{5s}{10} - \frac{5}{10} \cdot s = 0$